Data Structure Lab

BSAI and BSDS (III Evening)

Lab_08

Tree

New Lab Tasks

Instructor:

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Lab Tasks

INSTRUCTIONS:

- Add me as a collaborator: email: - sharifali.aulecturer@gmail.com.
- Clone your created repository: `data_structure_lab`.
- Inside the repository, create a **new folder**: `Lab_08_II_tree`
- **Complete all lab tasks** assigned in Lab task file.
- **Push your completed work** back to **GitHub**.
- **Make a Word or PDF file** and add your **screenshots** as you know better:
    - Take **screenshots** of your **code** and **the output** for each task.
    1. Include your code
    2. Include screenshots of outputs
    3. Add your **GitHub repository URL**
- **Upload the Word/PDF file to GCR** for submission

Lab Task

1. University Department Hierarchy — Binary Tree Creation

Scenario: A university wants to represent its department hierarchy, showing the Head of Department (HOD), coordinators, and faculty members in a tree structure.

Task: Create a binary tree using a `Node` class. Store the names of at least seven department members and connect them according to the department hierarchy.

Requirements:

- Create nodes using a class.
- Store the name of each department member.
- Connect parent and child nodes.
- Display each member's name along with their left and right children.

2. Online Shopping Categories — BFS Traversal

Scenario: An online shopping website organizes its products into different categories and subcategories.

Task: Create a binary tree containing at least seven product categories, such as Electronics, Clothing, Mobiles, Laptops, Men, Women, and Accessories. Display all categories using Breadth-First Traversal (BFS).

Requirements:

- Create a binary tree using a `Node` class.
- Use a `queue` to perform BFS.
- Visit nodes level by level.
- Display all product categories in BFS order.

3. Computer File Explorer — DFS Traversal

Scenario: A computer stores folders and subfolders. A user wants to explore folders by going deep into one folder before moving to another.

Task: Create a binary tree representing computer folders. Store at least seven folder names in the tree and display them using all three Depth-First Traversal (DFS) methods.

Requirements:

- Create a binary tree using a `Node` class.
- Store folder names in each node.
- Implement In-order, Pre-order, and Post-order traversal.
- Display folder names using all three traversal methods.
- Identify the visiting order of each traversal.

4. Hospital Appointment System — AVL Tree

Scenario: A hospital maintains patient appointment token numbers and wants to keep them organized for efficient searching.

Task: Create an AVL Tree and insert the appointment token numbers `15, 25, 35, 45, 55, 65, 75`. The tree should automatically balance itself whenever a new token number is inserted.

Requirements:

- Create an AVL Tree using a `Node` class.
- Store appointment token numbers in each node.
- Implement insertion according to BST rules.
- Calculate the height and balance factor of each node.
- Implement Left Rotation and Right Rotation.
- Handle all four rotation cases (LL, RR, LR, RL).
- Display appointment token numbers using In-order Traversal.
- Display the final balanced tree structure.

